CEBU INSTITUTE OF TECHNOLOGY UNIVERSITY

COLLEGE OF COMPUTER STUDIES

Software Requirements Specifications

for

WILDCAT WELFARE

Change History

Date	Revision	Description
05/06/2026	1.0	Initial Draft

Table of Contents

Change History **2**
Table of Contents **3**
1. Introduction **4**
1.1. Purpose 4
1.2. Scope 4
1.3. Definitions, Acronyms and Abbreviations 4
1.4. References 4
2. Overall Description **5**
2.1. Product perspective 5
2.2. User characteristics 5
2.4. Constraints 5
2.5. Assumptions and dependencies 6
3. Specific Requirements **7**
3.1. External interface requirements 7
3.1.1. Hardware interfaces *7*
3.1.2. Software interfaces *7*
3.1.3. Communications interfaces *7*
3.2. Functional requirements 7
Module 1 *7*
Module 2 *8*
3.4 Non-functional requirements 8
Performance *8*
Security *8*
Reliability *8*

1. Introduction

1.1. Purpose

The purpose of the Wildcat Welfare Portal is to digitally operate the current Animal Welfare Program needed at Cebu Institute of Technology - University (CIT-U). This system is a web-based initiative to connect the program's fragmented paper record keeping into one consolidated system under one governance structure.

The purpose of this document is to provide all stakeholders, (Development Team, Task Force Bruno, Medical-Dental Clinic Personnel, Academic Evaluators) with an easily understood common goal in moving forward with implementing the Wildcat Welfare Portal, outlining its key functionalities (i.e., QR coded pet profile, medical incident reporting for pets, adoption pipeline for pets, resource utilization), and creating a framework of quantifiable performance factors/metrics that will be used to determine successful implementation of the system.

1.2. Scope

The Wildcat Welfare Portal is a web-based platform created specifically to manage free-ranging dogs and cats located within the grounds of CIT-U University. This system will be used by Task Force Bruno (i.e., MDC, PCO, and SSD staff), potential adopters, student volunteers, and members of the surrounding university community.

- The system consists of 5 primary functionalities:
- QR Coded Pet Profiles: Digitized identification system that allows for instant access to a campus pet’s vaccination record, health history, and behavior notes by scanning the QR code.
- MDC Medical Dashboard: Restricted to clinic staff who log, manage, and track the medical records of pet-related visits, vaccination reminders, and report cases of bites or scratches.
- Community Newsfeed and Reporting Hub: A “live” module for reporting animal sightings outside the campus, posting missing pet alerts, and posting updates on animal welfare.
- Digital Adoption Pipeline: An integrated adoption application system that allows end-to-end tracking of how to rehome pets from their profile page.
- Resource and Donation Tracker: “live” inventory tracking dashboard that has an optional online payment option that tracks and monitors the distribution of food and medical supplies.

Visitors may not bring their personal pets onto campus, nor will animals located outside CIT premises be included in this system. The Task Force Bruno will continue to be responsible for the physical vetting of all applicants to adopt animals from the MDC; this will not be a part of the MDC's clinical operations, but will serve as a supplemental digital layer for documentation purposes only.

1.3. Definitions, Acronyms and Abbreviations

Term/Acronym	Definition
CIT-U	Cebu Institute of Technology - University
Task Force Bruno	The CIT-U committee responsible for the welfare, monitoring, and management of campus animals
MDC	Medical-Dental Clinic; the CIT-U health facility responsible for logging and treating animal bite/scratch incidents
PCO	Property and Campus Operations office; involved in campus animal monitoring and relocation
SSD	Student Services Division; a stakeholder unit involved in campus pet welfare advocacy
CNVR	Catch, Neuter, Vaccinate, and Return; a humane stray animal population control framework adopted by Task Force Bruno
QR Code	Quick Response Code; a machine-readable matrix barcode used in this system for instant pet profile retrieval
PEP	Post-Exposure Prophylaxis; the medical protocol administered after potential rabies exposure from an animal bite or scratch
SRS	Software Requirements Specification; this document
MVP	Minimum Viable Product; the initial deployable version of the system containing core features
SUS	System Usability Scale; a standardized usability evaluation tool. A score of ≥ 80 is the target for this system
Lawganan	A Visayan/Cebuano term referring to a designated feeding area or feeding territory for campus animals
Agile Scrum	An iterative software development methodology used in this project, organized into two-week sprint cycles
University Pet	A campus animal that is recognized, registered, and cared for under the CIT-U community and Task

	Force Bruno
Stray	An unregistered animal that has entered the campus premises and has not yet been assessed by Task Force Bruno

1.4. References

- International Organization for Standardization. (2011). Systems and software engineering — Systems and software Quality Requirements and Evaluation (SQuaRE) — System and software quality models (ISO/IEC Standard No. 25010:2011). https://www.iso.org/standard/35733.html
- Macarulay, P. M. M., Lanticse, V. C. B., Ubas, R. M., Estillore, E. L. M., & Tiu, J. E. (2026). The Wildcat Welfare Portal: Fostering a pet-friendly institution through digital governance and community advocacy. Cebu Institute of Technology – University.
- Medical-Dental Clinic. (2026). Proposal for policy/guideline on promoting a pet-friendly and responsible institution. Cebu Institute of Technology – University.
- Task Force Bruno. (2026, February 27). Minutes of the meeting: Proceedings for the 2026 plans for the campus Wildcats and Dogs. Cebu Institute of Technology – University.

2. Overall Description

2.1. Product perspective

The Wildcat Welfare Portal is a standalone, web-based management system designed to replace the fragmented paper-based record-keeping currently used by CIT-U's Task Force Bruno. The system operates as a centralized digital governance layer that interfaces with the daily operations of the Medical-Dental Clinic (MDC).

The system follows a client-server architecture:

- **Frontend:** A mobile-responsive web interface accessible to students and staff via smartphones for on-site QR code scanning and sighting reports.
- **Backend:** A secure administrative dashboard for MDC personnel to manage sensitive medical data and rabies exposure records.
- **Database:** A centralized repository for pet health histories, adoption logs, and community newsfeed data.

2.2. User characteristics

The system serves four primary user classes with varying levels of technical expertise and access rights:

- **Task Force Admins (MDC/PCO/SSD):** These are the primary decision-makers who require full access to medical dashboards, adoption vetting, and resource inventory. They possess moderate technical proficiency.
- **Student Volunteers/Moderators:** Highly tech-savvy students who assist in managing the community newsfeed, reporting sightings, and updating pet profiles.
- **University Community (Students/Faculty/Staff):** General users who primarily use the system to scan QR codes for pet identification and report sightings. They are expected to be highly proficient with mobile web browsing.
- **Potential Adopters:** Internal or external stakeholders who interact with the system specifically to view pet profiles and submit digital adoption applications.

2.4. Constraints

The development and operation of the Wildcat Welfare Portal are subject to the following constraints:

- **Connectivity:** The system is entirely dependent on a stable campus Wi-Fi or mobile data connection for real-time QR scanning and incident reporting.
- **Geographic Limitation:** The system scope is strictly limited to the CIT-U University Premises; no active GPS or geolocation tracking of animals outside campus is supported.
- **Regulatory Compliance:** Medical data handling must comply with university health privacy standards and local rabies protocols (RA 9482) regarding the reporting of bite incidents.
- **Hardware Limitation:** The system must remain lightweight enough to be functional on mid-range mobile devices used by security personnel and students for scanning.

2.5. Assumptions and dependencies

The Wildcat Welfare Portal operates under several assumptions and dependencies necessary for the successful implementation and operation of the system.

Assumptions

- It is assumed that all authorized users, including Task Force Bruno personnel, student volunteers, and MDC staff, possess basic digital literacy and are capable of navigating a web-based system using desktop or mobile devices.
- It is assumed that users will have access to stable internet connectivity through the CIT-U campus Wi-Fi network or mobile data services when accessing the platform.
- It is assumed that smartphones used for QR code scanning have functional cameras and updated web browsers capable of rendering of responsive webpage
- It is assumed that Task Force Bruno and MDC personnel will regularly update pet profiles, medical records, inventory logs, and adoption statuses to ensure data accuracy and reliability.
- It is assumed that the university administration will continue supporting the implementation of the CNVR (Catch, Neuter, Vaccinate, and Return) framework and related campus animal welfare initiatives.
- It is assumed that users will comply with university policies regarding responsible usage of the system and truthful submission of reports and adoption applications.

Dependencies

- The system depends on the availability and stability of the university's network infrastructure, including campus Wi-Fi and internet services.
- The system depends on a web hosting server and database management system capable of securely storing and processing application data.
- The system depends on third-party services such as email services (SMTP) for automated notifications and optional online payment gateways for donation processing.
- The QR Code identification feature depends on the availability and readability of printed or digitally generated QR codes attached to recognized university pets.
- The system depends on modern web browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari for proper functionality and compatibility.

The development of the system depends on the continued collaboration and approval of Task Force Bruno, MDC personnel, and university stakeholders for requirements validation and operational policies.

3. Specific Requirements

3.1. External interface requirements

The Wildcat Welfare Portal interacts with several hardware and software components to support the system's web-based operations, QR code identification processes, medical documentation, and community engagement features.

3.1.1. Hardware interfaces

The Wildcat Welfare Portal shall support the following hardware components:

- Smartphones and tablets with built-in cameras for scanning QR codes linked to pet profiles.
- Desktop and laptop computers used by Task Force Bruno and MDC personnel for accessing the administrative dashboard.
- Server hardware for storing pet records, medical logs, adoption applications, and community reports.
- Network devices such as routers and wireless access points to provide internet connectivity across the campus.
- QR code print materials attached to designated university pet areas for identification purposes

3.1.2. Software interfaces

The Wildcat Welfare Portal shall interface with the following software components:

- Web browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari for accessing the system.
- A database management system (e.g., MySQL) for storing user accounts, pet records, medical history, and adoption data.
- An email service for sending notifications such as vaccination reminders and adoption updates.
- QR code generation software or libraries for creating unique QR codes linked to pet profiles.
- Backend web technologies and REST APIs for handling communication between the frontend interface and the server.
- SSL/TLS encryption protocols to secure sensitive information during data transmission.

3.1.3. Communications interfaces

The Wildcat Welfare Portal relies on the following communication interfaces to support its web-based operations:

- HTTP/HTTPS Protocol: All client-server communication between the web frontend and the backend server shall be conducted over HTTPS (TLS 1.2 or higher) to ensure encrypted data transmission, particularly for medical records and adoption application data.

- RESTful API: The system shall expose and consume REST APIs for communication between the frontend interface and backend services (e.g., retrieving pet profiles upon QR code scan, submitting sighting reports to the Newsfeed, and processing adoption applications).

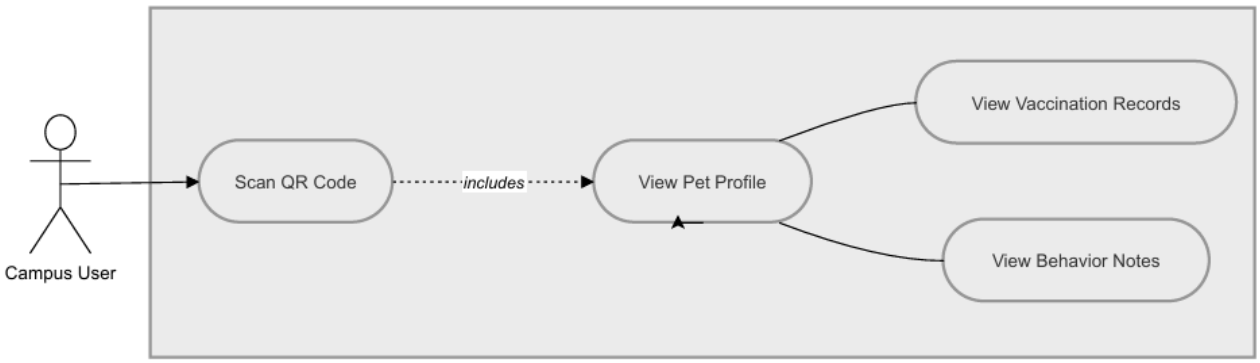
- QR Code Communication: Upon scanning a QR code, the mobile browser shall initiate an HTTPS GET request to the backend server to retrieve the corresponding pet profile. The QR code shall encode a unique URL pointing to the pet's profile endpoint.
- Campus Wi-Fi / Mobile Data: The system depends on the CIT-U campus Wi-Fi network or mobile data (3G/4G/5G) as the underlying transport network. No offline or peer-to-peer communication mode is supported.
- Email Notification Interface: The system shall interface with an SMTP-based email service to send automated notifications, such as adoption status updates, vaccination reminders, and missing pet alerts to registered users.
- SSL/TLS Encryption: All data transfers involving personally identifiable information (PII) — including adopter details and bite incident reports — must be encrypted using SSL/TLS protocols, in compliance with the system's data integrity requirements.
- Session Token Communication: Authentication and role-based access control shall be maintained through secure session tokens (e.g., JWT) transmitted via HTTPS headers between the client and server for all authenticated requests.

3.2. Functional requirements

Module 1

1.1: QR-Coded Profile Access

- *Use Case Diagram*

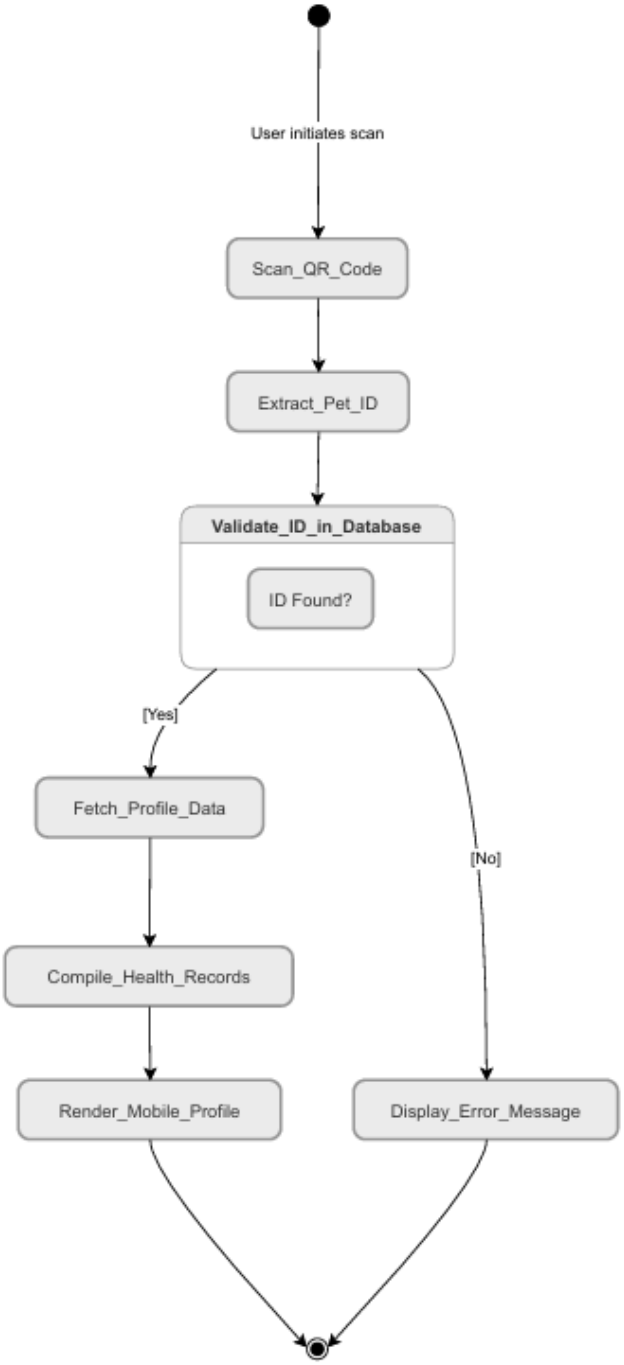


- *Use Case Description*

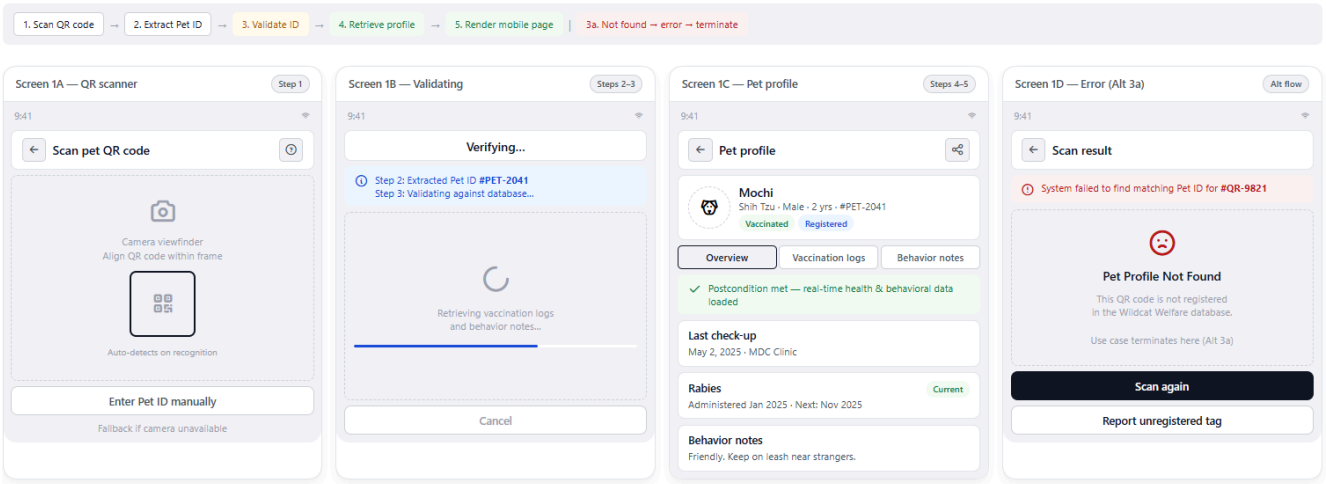
Use Case ID	**UC-1.1**
Use Case Name	QR-Coded Profile Access
Primary Actor	Campus User (Student, Faculty, or Staff)
Preconditions	The user has a smartphone with camera functionality. The scanned QR code is registered in the database.
Main Success Scenario	1. The User scans the physical QR code attached to the animal. 2. The System extracts the unique Pet ID from the QR code 3. The System validates the Pet ID against the database. 4. The System retrieves the associated pet profile (vaccination logs, behavior notes). 5. The System renders the mobile-responsive pet identification page.

Alternative Flows	**3a. Invalid QR Code:** 1. The System fails to find a matching Pet ID. 2. The System displays an error: "Pet Profile Not Found." 3. The use case terminates.
Postconditions	The user successfully views the real-time health and behavioral data of the campus pet.

- *Activity Diagram*

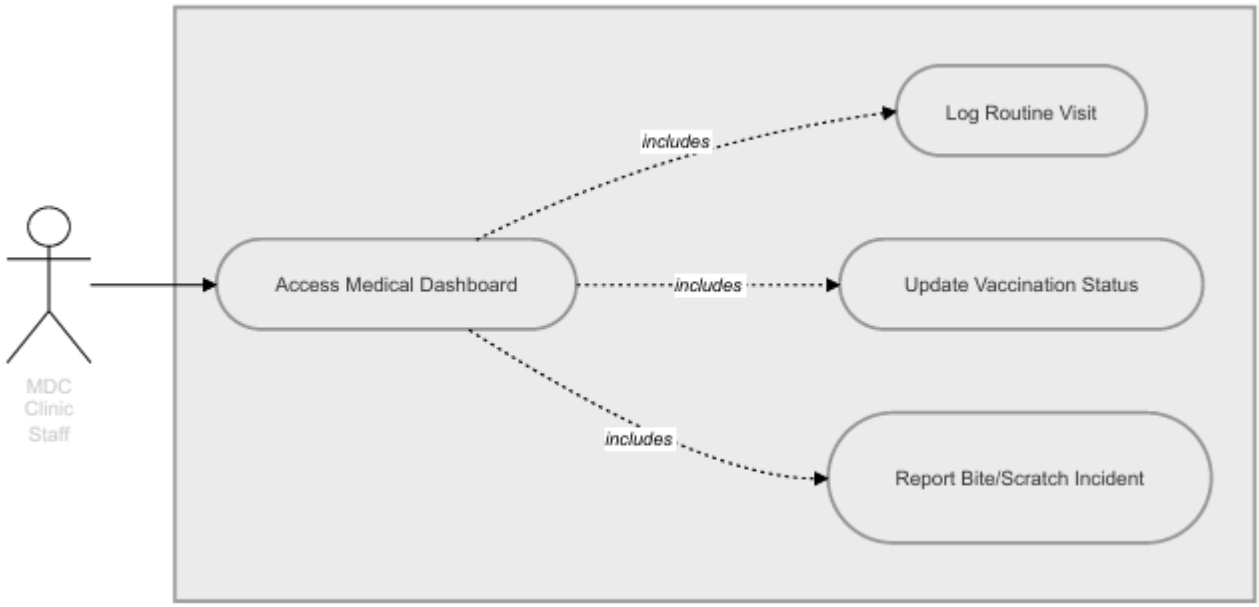


- *Wireframe*



1.2: MDC Medical Dashboard

- *Use Case Diagram*

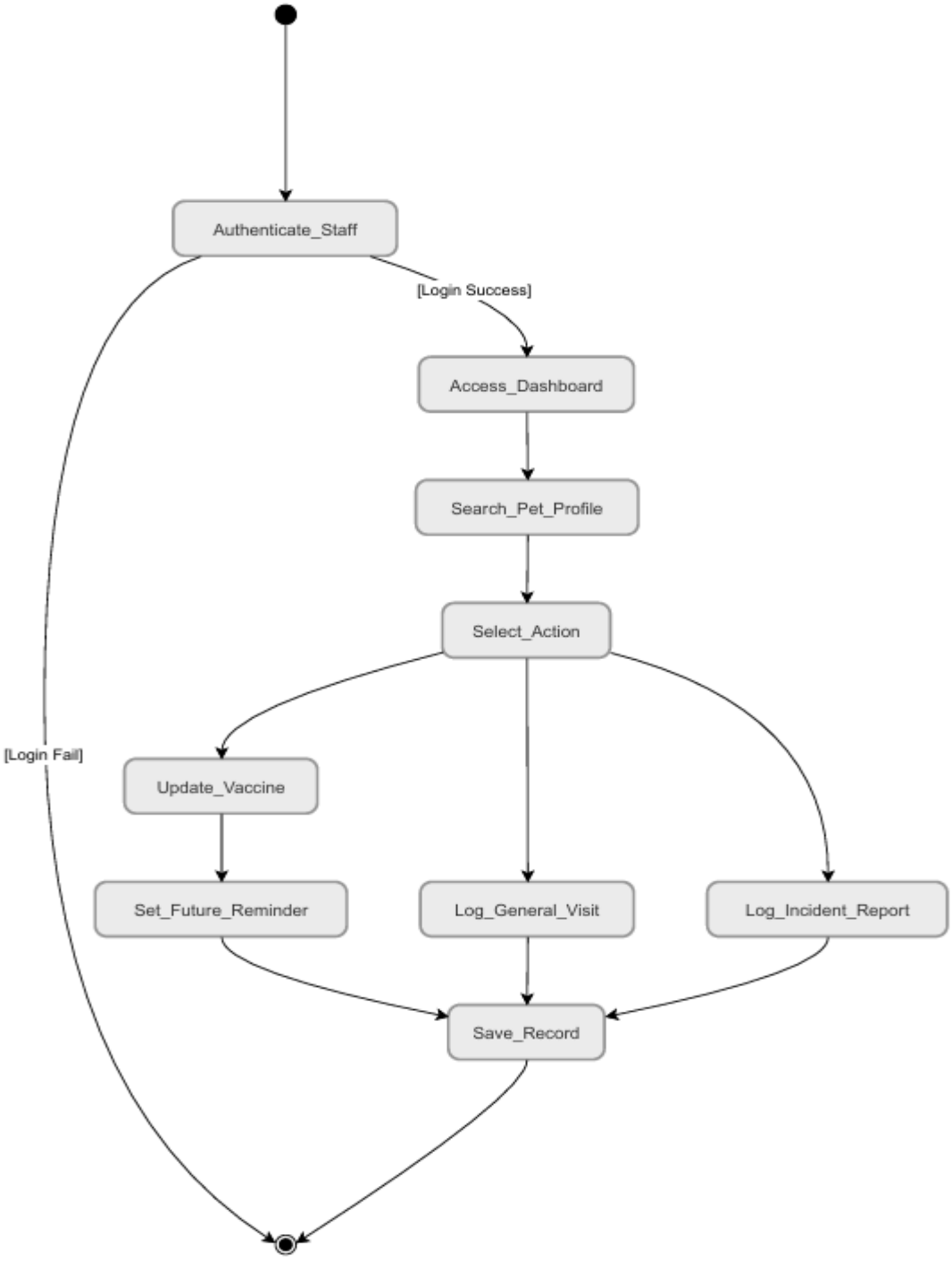


- *Use Case Description*

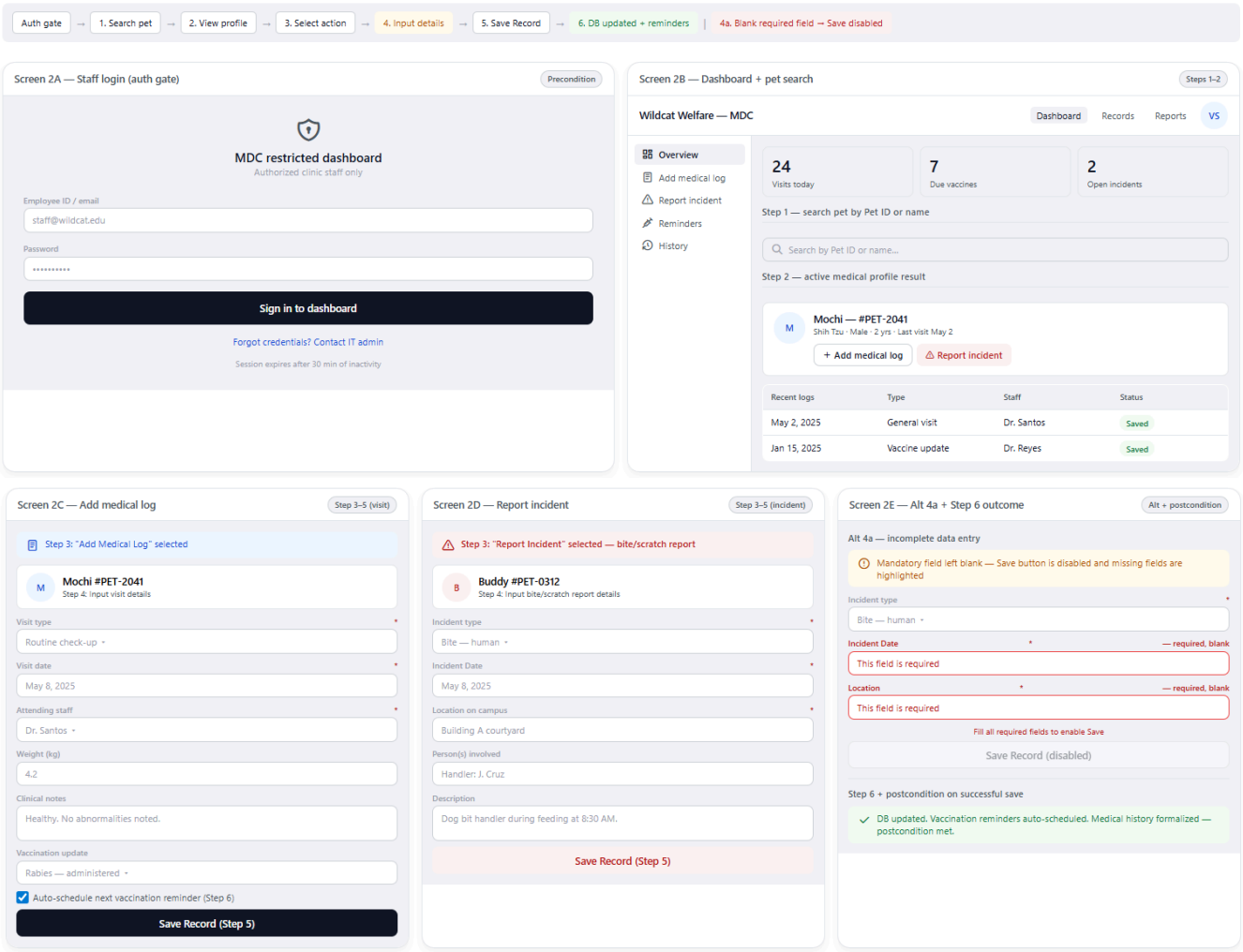
Use Case ID	**UC-1.2**
Use Case Name	Log Medical Records and Incidents
Primary Actor	MDC Clinic Staff
Preconditions	The staff member is authenticated and logged into the restricted medical dashboard.
Main Success Scenario	1. The Staff searches for a specific pet using the Pet ID or Name. 2. The System displays the active medical profile. 3. The Staff selects "Add Medical Log" or "Report Incident". 4. The Staff inputs visit details, vaccination updates, or bite/scratch reports. 5. The Staff clicks "Save Record". 6. The System updates the database and automatically schedules future vaccination reminders if applicable.

Alternative Flows	**4a. Incomplete Data Entry:** 1. The Staff leaves mandatory fields (e.g., Incident Date) blank. 2. The System disables the "Save" button and highlights missing fields.
Postconditions	The animal's medical history is updated, and the campus clinic has a formalized record of the visit or incident.

- *Activity Diagram*



- *Wireframe*

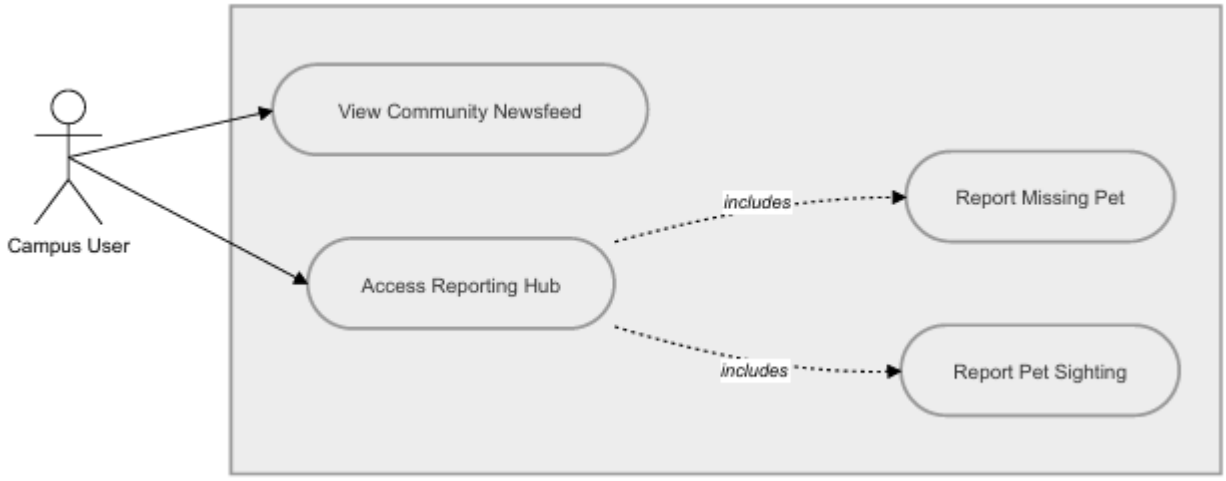

Auth gate
1. Search pet
2. View profile
3. Select action
4. Input details
5. Save Record
6. DB updated + reminders
4a. Blank required field → Save disabled
Screen 2A — Staff login (auth gate)
Precondition
MDC restricted dashboard
Authorized clinic staff only
Employee ID / email
staff@wildcat.edu
Password
Sign in to dashboard
Forgot credentials? Contact IT admin
Session expires after 30 min of inactivity
Screen 2B — Dashboard + pet search
Steps 1–2
Wildcat Welfare — MDC
Dashboard
Records
Reports
VS
Overview
Add medical log
Report incident
Reminders
History
24
Visits today
7
Due vaccines
2
Open incidents
Step 1 — search pet by Pet ID or name
Search by Pet ID or name...
Step 2 — active medical profile result
Mochi — #PET-2041
Shih Tzu · Male · 2 yrs · Last visit May 2
+ Add medical log
Report incident
Recent logs
Type
Staff
Status
May 2, 2025
General visit
Dr. Santos
Saved
Jan 15, 2025
Vaccine update
Dr. Reyes
Saved
Screen 2C — Add medical log
Step 3–5 (visit)
Step 3: "Add Medical Log" selected
Mochi #PET-2041
Step 4: Input visit details
Visit type
Routine check-up
Visit date
May 8, 2025
Attending staff
Dr. Santos
Weight (kg)
4.2
Clinical notes
Healthy. No abnormalities noted.
Vaccination update
Rabies — administered
Auto-schedule next vaccination reminder (Step 6)
Save Record (Step 5)
Screen 2D — Report incident
Step 3–5 (incident)
Step 3: "Report Incident" selected — bite/scratch report
Buddy #PET-0312
Step 4: Input bite/scratch report details
Incident type
Bite — human
Incident Date
May 8, 2025
Location on campus
Building A courtyard
Person(s) involved
Handler: J. Cruz
Description
Dog bit handler during feeding at 8:30 AM.
Save Record (Step 5)
Screen 2E — Alt 4a + Step 6 outcome
Alt + postcondition
Alt 4a — incomplete data entry
Mandatory field left blank — Save button is disabled and missing fields are highlighted
Incident type
Bite — human
Incident Date
— required, blank
This field is required
Location
— required, blank
This field is required
Fill all required fields to enable Save
Save Record (disabled)
Step 6 + postcondition on successful save
DB updated. Vaccination reminders auto-scheduled. Medical history formalized — postcondition met.

. . .

Module 2

2.1: Community Newsfeed and Reporting Hub

- *Use Case Diagram*



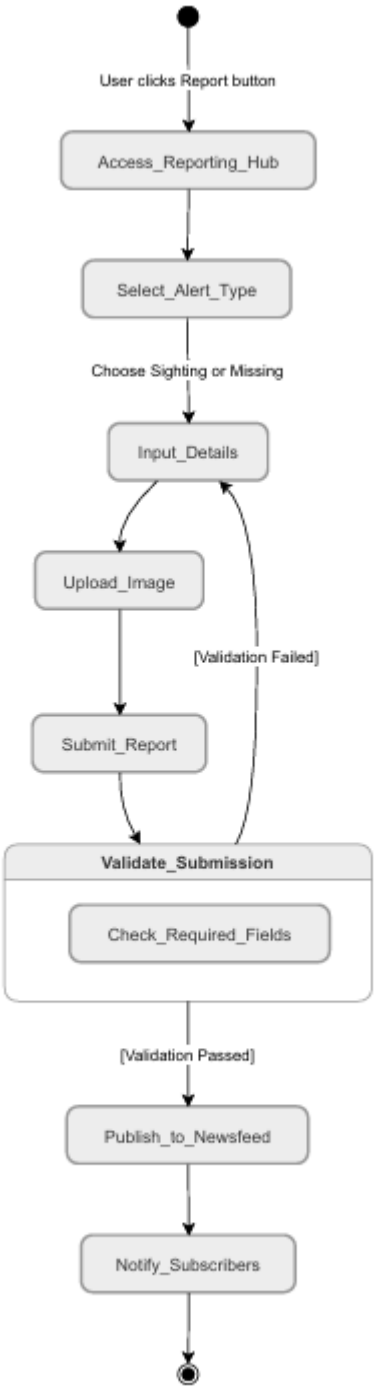
- *Use Case Description*

Use Case ID	**UC-2.1**
Use Case Name	Post Community Sighting or Missing Alert
Primary Actor	Community Member
Preconditions	The user is accessing the Community Newsfeed module.

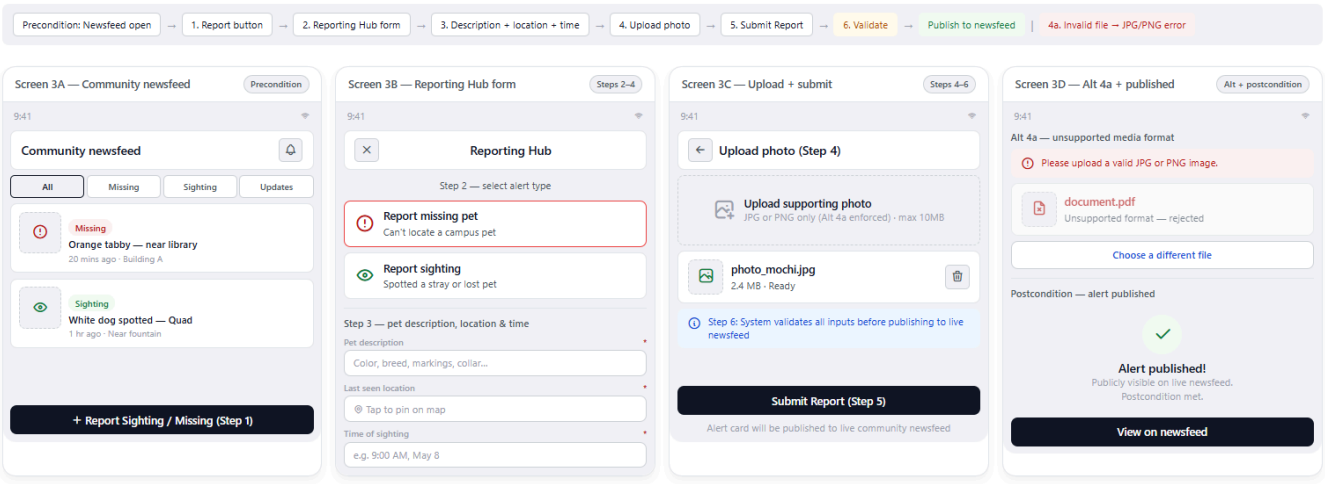
Main Success Scenario	1. The User clicks the "Report Sighting/Missing" button. 2. The System presents the Reporting Hub form. 3. The User inputs the pet description, last seen location, and time. 4. The User uploads a supporting photo. 5. The User clicks "Submit Report". 6. The System validates the inputs and publishes the alert card to the live "Community Newsfeed".
Alternative Flows	**4a. Unsupported Media Format:**

	1. The User attempts to upload a file type not supported by the system. 2. The System displays an error: "Please upload a valid JPG or PNG image."
Postconditions	A new alert is publicly visible on the live newsfeed to inform the campus community.

- *Activity Diagram*

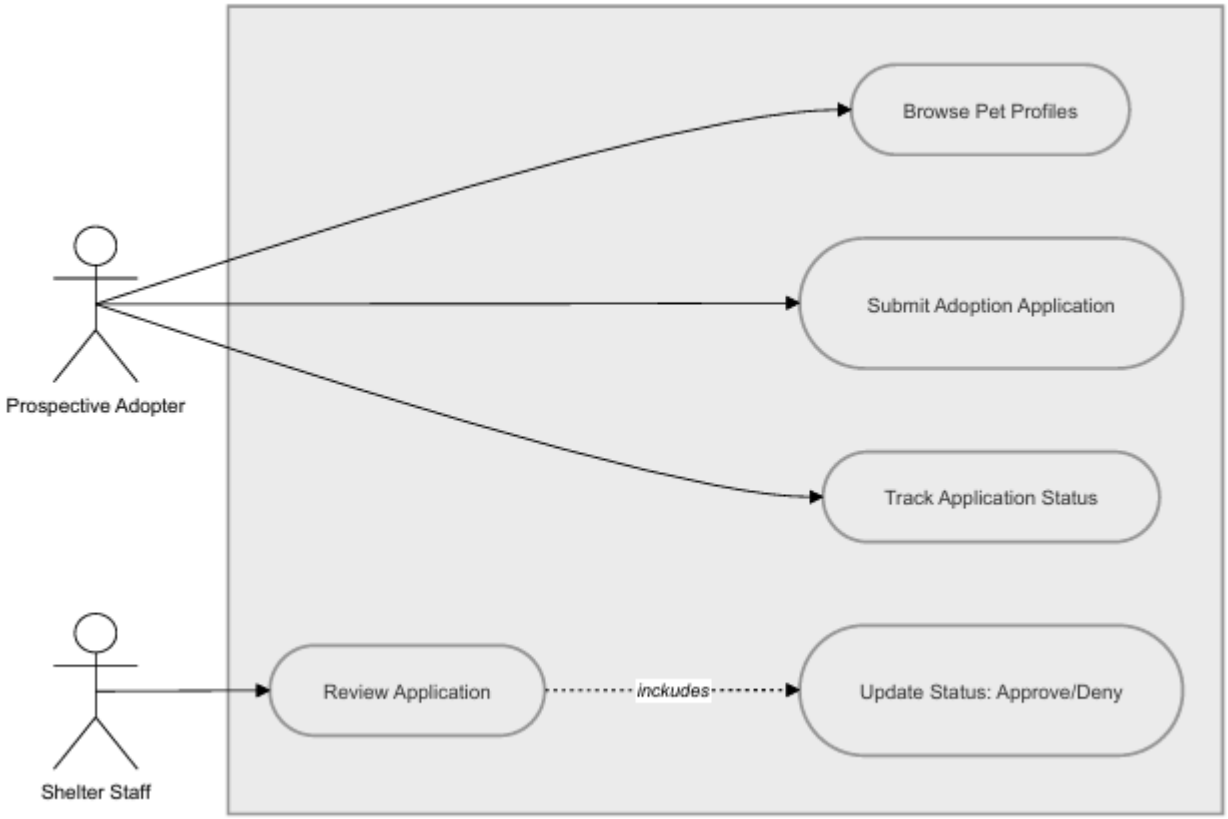


- *Wireframe*



2.2: Digital Adoption Pipeline

- *Use Case Diagram*

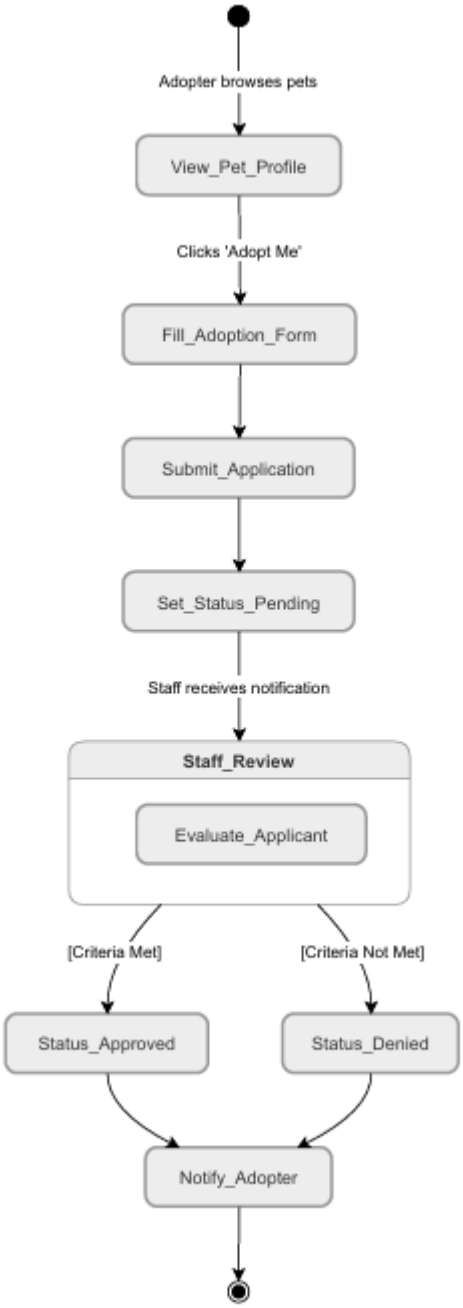


- *Use Case Description*

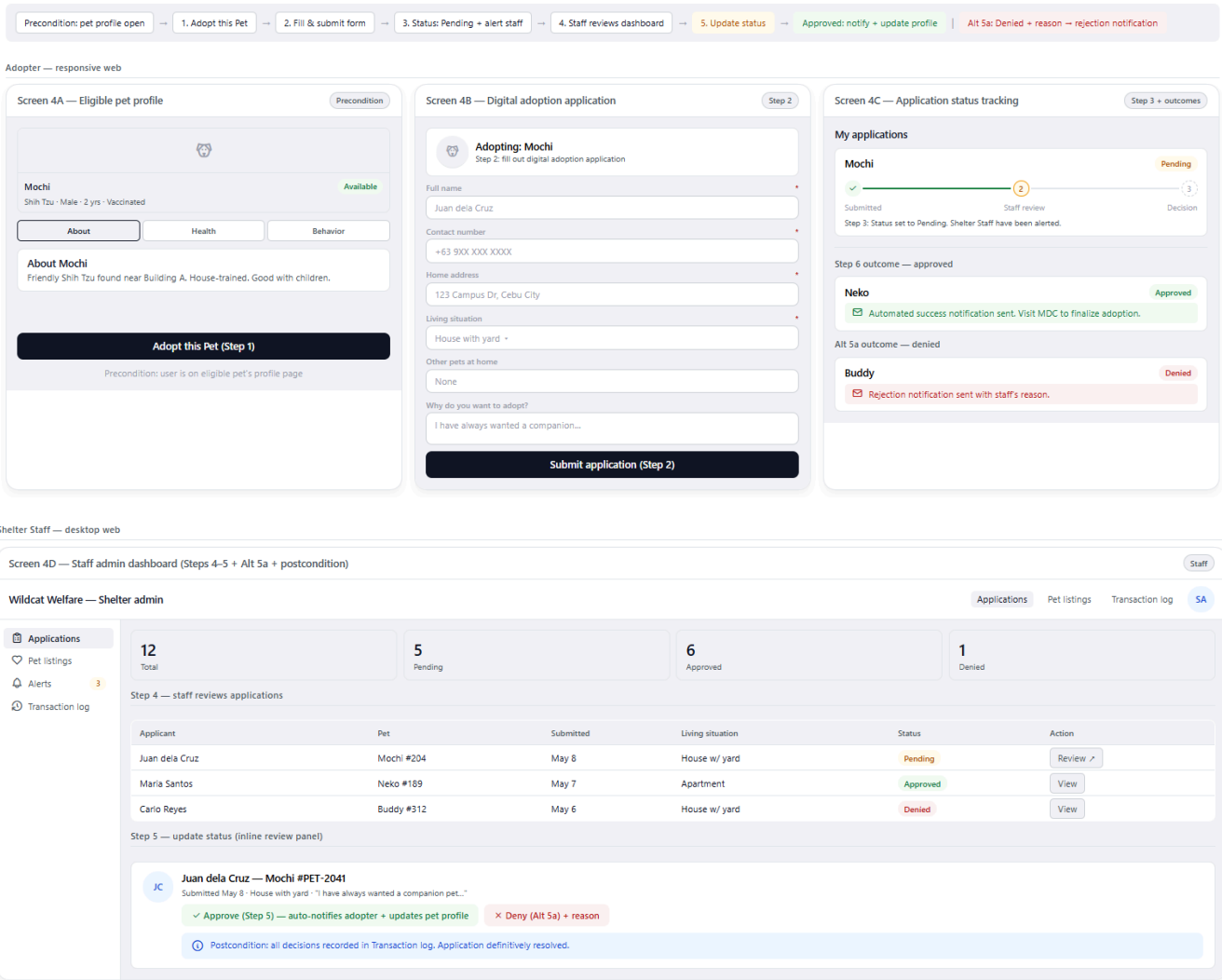
Use Case ID	**UC-2.2**
Use Case Name	Submit and Process Adoption Application
Primary Actor	Prospective Adopter, Shelter Staff
Preconditions	The Adopter is viewing an eligible pet's profile page.
Main Success Scenario	1. The Adopter clicks "Adopt this Pet". 2. The Adopter fills out and submits the digital adoption application. 3. The System sets the application status to "Pending" and alerts Shelter Staff. 4. Staff accesses the admin dashboard and reviews the application.

	5. Staff updates the status to "Approved". 6. The System sends an automated success notification to the Adopter and updates the pet's profile status.
Alternative Flows	**5a. Application Denied:** 1. Staff updates the status to "Denied" with a specified reason. 2. The System sends a rejection notification to the Adopter.
Postconditions	The adoption application is definitively resolved, and the system records the transaction history.

- *Activity Diagram*



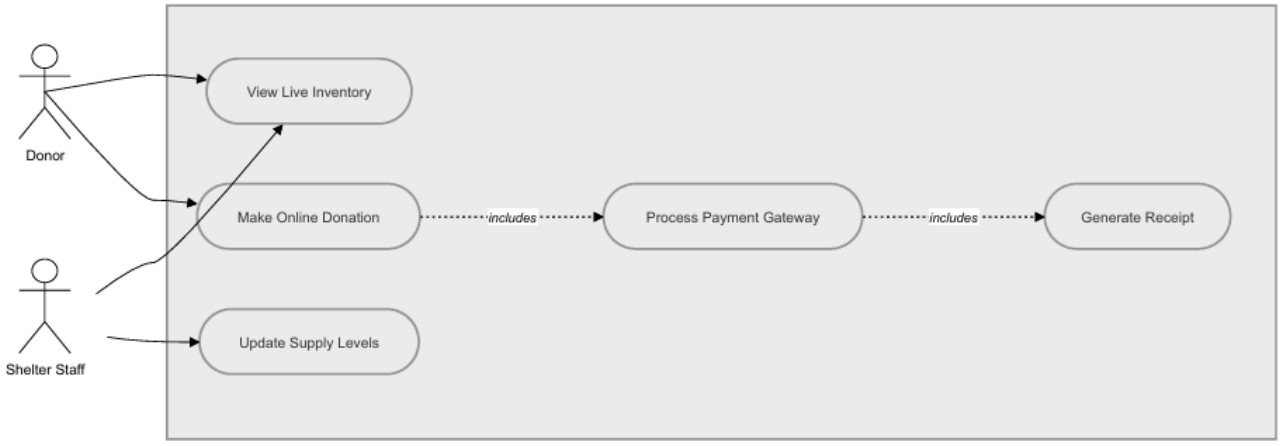
- *Wireframe*


Precondition: pet profile open
1. Adopt this Pet
2. Fill & submit form
3. Status: Pending + alert staff
4. Staff reviews dashboard
5. Update status
Approved: notify + update profile
Alt 5a: Denied + reason → rejection notification
Adopter — responsive web
Screen 4A — Eligible pet profile
Precondition
Mochi
Available
Shih Tzu · Male · 2 yrs · Vaccinated
About
Health
Behavior
About Mochi
Friendly Shih Tzu found near Building A. House-trained. Good with children.
Adopt this Pet (Step 1)
Precondition: user is on eligible pet's profile page
Screen 4B — Digital adoption application
Step 2
Adopting: Mochi
Step 2: fill out digital adoption application
Full name
Juan dela Cruz
Contact number
+63 9XX XXX XXXX
Home address
123 Campus Dr, Cebu City
Living situation
House with yard
Other pets at home
None
Why do you want to adopt?
I have always wanted a companion...
Submit application (Step 2)
Screen 4C — Application status tracking
Step 3 + outcomes
My applications
Mochi
Pending
Submitted
Staff review
Decision
Step 3: Status set to Pending. Shelter Staff have been alerted.
Step 6 outcome — approved
Neko
Approved
Automated success notification sent. Visit MDC to finalize adoption.
Alt 5a outcome — denied
Buddy
Denied
Rejection notification sent with staff's reason.
Shelter Staff — desktop web
Screen 4D — Staff admin dashboard (Steps 4–5 + Alt 5a + postcondition)
Staff
Wildcat Welfare — Shelter admin
Applications
Pet listings
Transaction log
SA
Applications
Pet listings
Alerts
3
Transaction log
12
Total
5
Pending
6
Approved
1
Denied
Step 4 — staff reviews applications
Applicant
Pet
Submitted
Living situation
Status
Action
Juan dela Cruz
Mochi #204
May 8
House w/ yard
Pending
Review ↗
Maria Santos
Neko #189
May 7
Apartment
Approved
View
Carlo Reyes
Buddy #312
May 6
House w/ yard
Denied
View
Step 5 — update status (inline review panel)
JC
Juan dela Cruz — Mochi #PET-2041
Submitted May 8 · House with yard · "I have always wanted a companion pet..."
✓ Approve (Step 5) — auto-notifies adopter + updates pet profile
× Deny (Alt 5a) + reason
Postcondition: all decisions recorded in Transaction log. Application definitively resolved.

Module 3

3: Resource & Donation Logistics

- *Use Case Diagram*



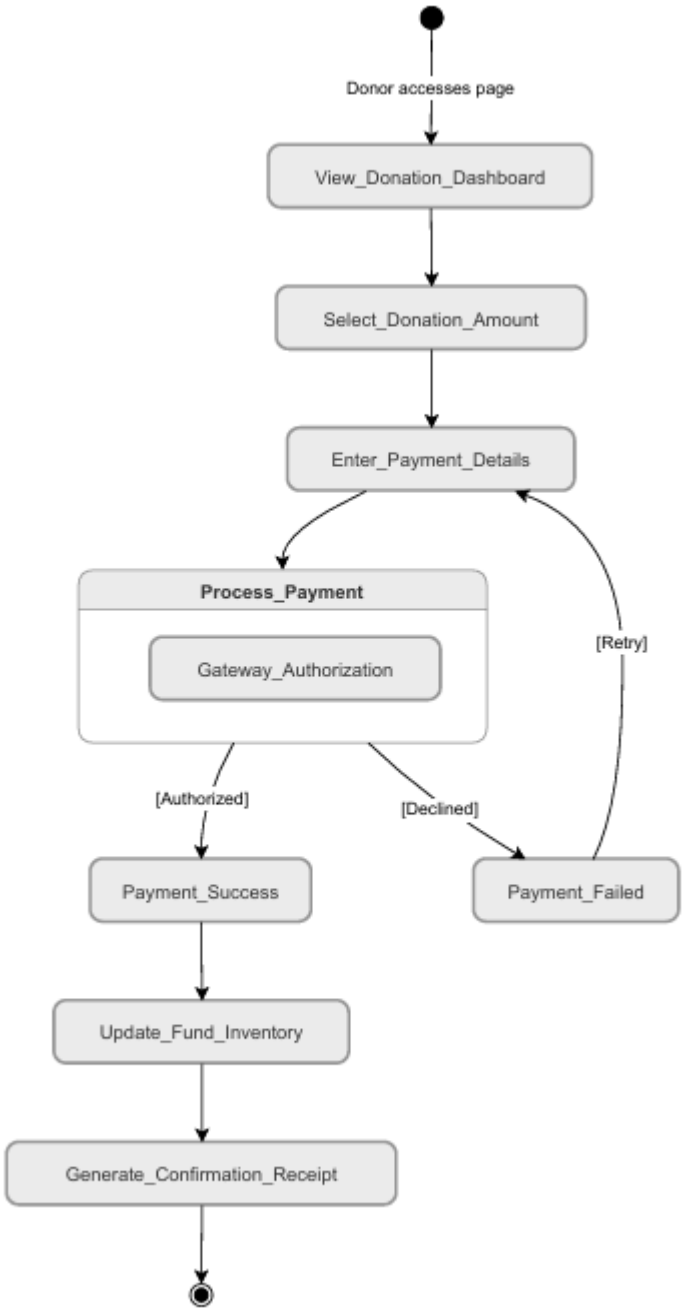
- *Use Case Description*

Use Case ID	**UC-3.1**
Use Case Name	Track Inventory and Process Online Donations
Primary Actor	Donor, Shelter Staff
Preconditions	The Donor has valid payment credentials; Staff has administrative access.

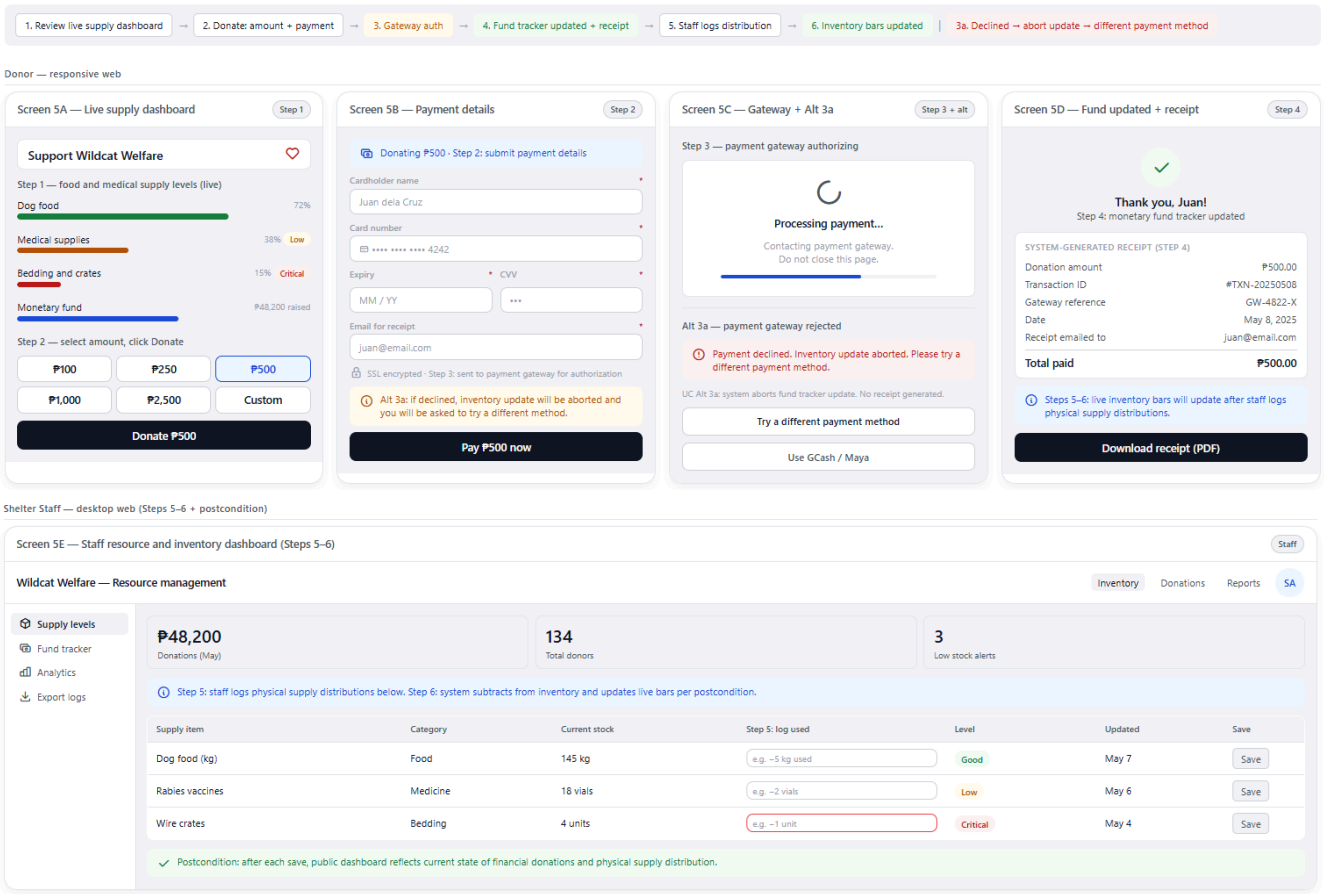
Main Success Scenario	1. The Donor reviews the live dashboard showing food and medical supply levels. 2. The Donor clicks "Donate", enters a monetary amount, and submits payment details. 3. The Payment Gateway authorizes the transaction. 4. The System automatically updates the monetary fund tracker and generates a receipt. 5. (Staff Action) Staff logs physical supply distributions (e.g., 5kg of dog food used). 6. The System updates the live inventory bars reflecting the subtracted supplies.

Alternative Flows	**3a. Payment Declined:** 1. The Payment Gateway rejects the transaction. 2. The System aborts the inventory update and prompts the Donor to try a different payment method.
Postconditions	The public dashboard accurately reflects the current state of both financial donations and physical supply distribution.

- *Activity Diagram*



- *Wireframe*


1. Review live supply dashboard
2. Donate: amount + payment
3. Gateway auth
4. Fund tracker updated + receipt
5. Staff logs distribution
6. Inventory bars updated
3a. Declined → abort update → different payment method
Donor — responsive web
Screen 5A — Live supply dashboard
Step 1
Support Wildcat Welfare
Step 1 — food and medical supply levels (live)
Dog food
72%
Medical supplies
38%
Low
Bedding and crates
15%
Critical
Monetary fund
₱48,200 raised
Step 2 — select amount, click Donate
₱100
₱250
₱500
₱1,000
₱2,500
Custom
Donate ₱500
Screen 5B — Payment details
Step 2
Donating ₱500 · Step 2: submit payment details
Cardholder name
Juan dela Cruz
Card number
•••• •••• •••• 4242
Expiry
MM / YY
CVV
Email for receipt
juan@email.com
SSL encrypted · Step 3: sent to payment gateway for authorization
Alt 3a: if declined, inventory update will be aborted and you will be asked to try a different method.
Pay ₱500 now
Screen 5C — Gateway + Alt 3a
Step 3 + alt
Step 3 — payment gateway authorizing
Processing payment...
Contacting payment gateway.
Do not close this page.
Alt 3a — payment gateway rejected
Payment declined. Inventory update aborted. Please try a different payment method.
UC Alt 3a: system aborts fund tracker update. No receipt generated.
Try a different payment method
Use GCash / Maya
Screen 5D — Fund updated + receipt
Step 4
Thank you, Juan!
Step 4: monetary fund tracker updated
SYSTEM-GENERATED RECEIPT (STEP 4)
Donation amount
₱500.00
Transaction ID
#TXN-20250508
Gateway reference
GW-4822-X
Date
May 8, 2025
Receipt emailed to
juan@email.com
Total paid
₱500.00
Steps 5–6: live inventory bars will update after staff logs physical supply distributions.
Download receipt (PDF)
Shelter Staff — desktop web (Steps 5–6 + postcondition)
Screen 5E — Staff resource and inventory dashboard (Steps 5–6)
Staff
Wildcat Welfare — Resource management
Inventory
Donations
Reports
SA
Supply levels
Fund tracker
Analytics
Export logs
₱48,200
Donations (May)
134
Total donors
3
Low stock alerts
Step 5: staff logs physical supply distributions below. Step 6: system subtracts from inventory and updates live bars per postcondition.
Supply item
Category
Current stock
Step 5: log used
Level
Updated
Save
Dog food (kg)
Food
145 kg
e.g. −5 kg used
Good
May 7
Save
Rabies vaccines
Medicine
18 vials
e.g. −2 vials
Low
May 6
Save
Wire crates
Bedding
4 units
e.g. −1 unit
Critical
May 4
Save
Postcondition: after each save, public dashboard reflects current state of financial donations and physical supply distribution.

3.4 Non-functional requirements

Performance

• **Identification Speed:** The system must retrieve and display a pet's digital profile from a scanned QR code in under 20 seconds.
• **Reporting Latency:** Community sighting reports must be visible on the Newsfeed within 5 seconds of submission.
• **Concurrency:** The system must support at least 15 simultaneous users scanning codes or viewing the newsfeed during peak campus hours without significant lag.

Security

• **Access Control:** Role-based access must be enforced; only authorized MDC personnel can view or edit sensitive medical and bite incident logs.
• **Data Integrity:** Adoption applications must be protected via SSL encryption to ensure the privacy of potential adopters' personal information.
• **Session Management:** Admin sessions must automatically time out after 15 minutes of inactivity to prevent unauthorized access at clinic terminals.

Reliability

• **Uptime:** The system must maintain 99% availability during university operating hours (7:00 AM – 9:00 PM).
• **Data Backup:** Automated daily backups of the medical and adoption databases must be performed to prevent data loss.
• **Error Handling:** The system must provide user-friendly error messages if a scan fails or if the user is outside the CIT-U network coverage area.